

SOFTWARE REQUIREMENTS SPECIFICATION

ALUMNI WEB APPLICATION

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01. Project Overview

The Alumni Portal is a comprehensive web-based application designed to connect past students, current professionals, and educational institutions including colleges, schools, and offices. This platform serves as a unified digital bridge that enables alumni to maintain lifelong connections with their alma mater and fellow graduates.

1.1 Purpose

The Alumni Web Application aims to create a centralized platform where alumni can register, share career updates, network with peers, and stay engaged with their institution. The system facilitates seamless communication between institutions and their graduates while providing administrators with powerful tools to manage and showcase alumni achievements.

1.2 Key Objectives

- Build a unified networking platform for alumni across different batches and departments
- Enable institutions to effectively manage and showcase alumni achievements and success stories
- Provide secure authentication mechanisms for both administrators and users
- Implement advanced search and filtering capabilities for finding alumni by profession, batch, organization, or location
- Create a responsive and user-friendly interface accessible across all devices
- Establish a secure and scalable foundation for future feature enhancements

1.3 Scope

The Alumni Portal encompasses three primary user roles with distinct functionalities. Administrators have comprehensive control over alumni data with full CRUD operations, user management, and approval workflows. Alumni users can view, search, and connect with fellow graduates through a secure login system. The public interface showcases alumni achievements and provides registration capabilities through institution-linked forms. The application supports multiple institutions, each with customized branding and dedicated data management.

02. Website Features - Customised Landing Modules

2.1 Home Page Module

The home page serves as the main entry point featuring a visually appealing layout with the following components:

- Dynamic Header with navigation menu (Home, Register, User Login, Dashboard, Logout)
- Hero section showcasing 'Our Distinguished Alumni' with engaging visuals
- Alumni network counter displaying total registered members
- Alumni profile cards with carousel navigation showing key details
- Employment status badges (Studying, Employed, Self-Employed)
- Profile information including name, profession, organization, and batch year
- Comprehensive footer with contact information and quick links

2.2 Registration Module

A comprehensive registration system designed for alumni including students, officemates, and professionals from various institutions:

- User-friendly multi-step registration form with clear field labels
- Photo upload functionality with file size validation (JPG/PNG/GIF supported)
- Login credentials section for username and email address creation
- Personal information fields (full name, contact details, current location)
- Professional details section (current position, company/organization)
- Academic information (institution name, department, graduation year)
- Employment status selection (Studying, Employed, Self-Employed)
- Password creation with confirmation field and strength indicator
- Real-time form validation with helpful error messages
- Required field indicators with asterisk notation

2.3 Admin Dashboard Module

A powerful administrative interface providing comprehensive alumni database management:

- Dashboard overview with key statistics and metrics
- Add New Alumni button with streamlined profile creation workflow
- Manage Alumni section with view, edit, and delete capabilities
- Manage Users functionality for account creation and permissions
- Statistics Overview displaying total alumni count by employment status
- Visual analytics with icons showing Employed, Self-Employed, and Studying counts
- Search and filter tools for efficient data management
- Bulk operations support for data import/export
- User credential management system for creating and resetting login passwords

03. Website Features - Alumni Portal

3.1 User Authentication System

Feature	Description
Admin Login	Secure authentication with username/email and password for administrative access
User Login	Alumni member login with view-only access to alumni directory
Password Security	Encrypted password storage using bcrypt with salt rounds for enhanced security
Session Management	JWT token-based authentication with configurable expiration times

3.2 Alumni Directory Features

Feature	Description
Profile Display	Comprehensive alumni profiles with photo, profession, organization, batch year, department, and location
Advanced Search	Multi-criteria search by name, profession, company, batch year, department, or location
Filter Options	Filter by employment status (Studying, Employed, Self-Employed), batch year, or department
Contact Feature	Direct contact options through email or integrated messaging system

3.3 Data Management Capabilities

Administrators possess comprehensive control over the alumni database:

- Create new alumni profiles with complete information capture
- View detailed alumni information with full profile history
- Update existing records including personal, academic, and professional details
- Delete alumni records with confirmation safeguards
- Approve or reject alumni registration requests
- Export alumni data in CSV/Excel format for reporting
- Import bulk alumni data with validation and error handling
- Create and manage user login credentials for alumni access
- Reset passwords and manage user account permissions

04. Website Navigation Structure: Front End - English

4.1 Public Interface Navigation

Page	URL Path	Description
Home	/home or /	Landing page displaying featured alumni profiles and network statistics
Register	/register	Alumni registration form for new member enrollment
User Login	/user/login	Login portal for registered alumni members
About Us	/about	Information about the institution and alumni network
Contact	/contact	Contact form and institution contact details

4.2 User Interface Navigation (Alumni Members)

Page	URL Path	Description
User Dashboard	/user/dashboard	Personal dashboard for logged-in alumni
Alumni Directory	/admin/manage	View-only access to complete alumni directory
Search Alumni	/ admin/manage	Advanced search and filter interface
My Profile	/user/profile	View personal profile and update password

05. Administrative Module

5.1 Admin Authentication

Secure administrator login with multi-factor authentication support. Admin credentials are managed through encrypted storage with role-based access control ensuring only authorized personnel can access administrative functions.

5.2 Dashboard Overview

The admin dashboard provides a comprehensive overview with:

- Real-time statistics on total alumni count
- Employment status breakdown (Employed, Self-Employed, Studying)
- Recent registration requests pending approval
- Quick action buttons for common tasks
- System notifications and alerts

5.3 Alumni Management Functions

5.3.1 Add New Alumni

Administrators can create new alumni profiles with comprehensive data input including personal details, academic information, professional background, and contact information. The system validates all inputs and ensures data integrity.

5.3.2 Manage Alumni

Complete CRUD operations for alumni records including view detailed profiles, edit any field with validation, delete records with confirmation, approve or reject registration requests, and bulk operations for efficiency.

5.3.3 Manage Users

User account management capabilities including create login credentials for alumni members, reset passwords when requested, manage user permissions and access levels, deactivate or suspend user accounts, and track login activity and user engagement.

5.4 Search and Filter Tools

Advanced search capabilities for administrators:

- Full-text search across all alumni fields
- Filter by employment status, batch year, department, location
- Sort results by various criteria (name, date, status)
- Export filtered results to Excel/CSV
- Save custom filter configurations

06. Website Navigation Structure: Back End - Admin Section

6.1 Admin Panel Navigation

Module	URL Path	Access Level
Admin Login	/admin/login	Public (authentication required)
Dashboard	/admin/dashboard	Admin Only
Add Alumni	/admin/alumni/add	Admin Only
Manage Alumni	/admin/alumni/manage	Admin Only
Edit Alumni	/admin/alumni/edit/:id	Admin Only
Delete Alumni	/admin/alumni/delete/:id	Admin Only
Manage Users	/admin/users/manage	Admin Only
Create User	/admin/users/create	Admin Only
Reports	/admin/reports	Admin Only
Settings	/admin/settings	Admin Only

6.2 API Endpoints (Backend)

Method	Endpoint	Description
POST	/api/admin/login	Admin authentication endpoint
GET	/api/alumni	Retrieve all alumni records
GET	/api/alumni/:id	Retrieve single alumni record
POST	/api/alumni	Create new alumni record
PUT	/api/alumni/:id	Update alumni record
DELETE	/api/alumni/:id	Delete alumni record
POST	/api/users/create	Create user login credentials
POST	/api/search	Search and filter alumni

07. Domain, Hosting & Other Requirements

7.1 Domain Requirements

Domain Name	Custom domain (e.g., alumni.institution.edu or alumninet.com)
SSL Certificate	Required for HTTPS (Let's Encrypt or commercial SSL)
DNS Management	Access to DNS settings for configuration

7.2 Hosting Specifications

Component	Specification
Server Type	VPS or Cloud Server (AWS, Digital Ocean, Vercel, or Railway)
CPU	Minimum 2 vCPU cores (recommended 4 cores for production)
RAM	Minimum 4GB (recommended 8GB for optimal performance)
Storage	50GB SSD storage minimum (scalable based on alumni count)
Bandwidth	Minimum 1TB/month data transfer
Operating System	Ubuntu 20.04 LTS or higher, CentOS 8, or similar Linux distribution
Database	MySQL 8.0 or higher (dedicated or managed service)

7.3 Additional Requirements

- Email service for notifications (SendGrid, AWS SES, or SMTP server)
- CDN for static assets and image delivery (Cloudflare or AWS CloudFront)
- Backup solution with daily automated backups
- Monitoring and alerting system (e.g., PM2, New Relic, or DataDog)
- File storage for alumni photos (local or cloud storage like AWS S3)

08. Technical Proposal

8.1 Technology Stack

8.1.1 Frontend Technologies

Framework	React.js with Vite build tool
UI Components	Material-UI or Tailwind CSS for responsive design
State Management	React Context API or Redux for complex state management
Routing	React Router v6 for client-side navigation
Form Handling	React Hook Form with Yup validation

8.1.2 Backend Technologies

Runtime	Node.js (LTS version 18.x or higher)
Framework	Express.js for RESTful API development
Authentication	JWT (JSON Web Tokens) for secure session management
Password Encryption	bcrypt with salt rounds for password hashing
File Upload	Multer middleware for handling multipart/form-data
Validation	express-validator for input sanitization and validation

8.1.3 Database

Database System	MySQL 8.0 relational database
ORM/Query Builder	Sequelize ORM or raw MySQL2 driver for direct queries
Connection Pooling	Configured for optimal performance with 10-20 connections

8.2 Database Schema Design

8.2.1 Alumni Table

Primary table storing comprehensive alumni information:

- id (INT, PRIMARY KEY, AUTO_INCREMENT)
- name (VARCHAR 255, NOT NULL)
- email (VARCHAR 255, UNIQUE, NOT NULL)
- phone (VARCHAR 20)
- profession (VARCHAR 100)
- company (VARCHAR 255)
- college (VARCHAR 255)
- department (VARCHAR 100)
- batch_year (INT)

- location (VARCHAR 255)
- status (ENUM: 'Studying', 'Employed', 'Self-Employed')
- photo (VARCHAR 500 - file path)
- created_at (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)
- updated_at (TIMESTAMP, ON UPDATE CURRENT_TIMESTAMP)

8.2.2 Users Table

Authentication and user management table:

- id (INT, PRIMARY KEY, AUTO_INCREMENT)
- username (VARCHAR 100, UNIQUE, NOT NULL)
- email (VARCHAR 255, UNIQUE, NOT NULL)
- password_hash (VARCHAR 255, NOT NULL)
- role (ENUM: 'admin', 'user', DEFAULT 'user')
- alumni_id (INT, FOREIGN KEY references alumni.id)
- last_login (TIMESTAMP)
- is_active (BOOLEAN, DEFAULT TRUE)
- created_at (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)

8.3 Security Measures

- Password hashing with bcrypt (minimum 10 salt rounds)
- JWT token-based authentication with secure HTTP-only cookies
- CORS (Cross-Origin Resource Sharing) configuration
- SQL injection prevention through parameterized queries
- XSS (Cross-Site Scripting) protection with input sanitization
- CSRF (Cross-Site Request Forgery) tokens for form submissions
- Rate limiting to prevent brute force attacks
- HTTPS enforcement with SSL/TLS certificate
- File upload validation (type, size, malware scanning)
- Environment variables for sensitive configuration

09. Proposal Confidentiality & Disclosure Clause

This Software Requirements Specification document contains confidential and proprietary information belonging to Sunsys Technologies Pvt Ltd. The information contained herein is provided solely for the purpose of evaluating our proposed Alumni Web Application solution and must be treated with strict confidentiality.

9.1 Confidential Information

Confidential information includes but is not limited to:

- Technical specifications and architecture details
- Development methodologies and processes
- Pricing information and cost estimates
- Implementation timelines and resource allocation
- Proprietary software solutions and algorithms
- Any other technical or business information marked as confidential

9.2 Recipient Obligations

The recipient of this document agrees to:

- Maintain confidentiality of all information contained in this document
- Not disclose any information to third parties without written consent
- Use the information solely for evaluation purposes
- Return or destroy all copies upon request or project completion
- Limit access to authorized personnel with legitimate need-to-know

9.3 Intellectual Property

All intellectual property rights including but not limited to software code, architecture designs, documentation, and methodologies described in this proposal remain the exclusive property of Sunsys Technologies Pvt Ltd. No license or rights are granted except as explicitly stated in a signed agreement.

9.4 Duration

The confidentiality obligations shall remain in effect for a period of three (3) years from the date of disclosure, or until the information becomes publicly available through no fault of the recipient, whichever occurs first.

10. Development Sprints & Milestones

10.1 Project Timeline Overview

The Alumni Web Application will be developed using an Agile methodology with 2-week sprint cycles. The complete project is estimated to be completed in 12-14 weeks, divided into 6-7 sprints with clear deliverables and milestones.

10.2 Sprint Breakdown

Sprint 1: Project Setup & Database Design (Weeks 1-2)

Deliverables	• Project repository setup and version control • Development environment configuration • Database schema design and implementation • Basic Express.js server setup with MySQL connection • Project documentation structure
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Sprint 2: Authentication System (Weeks 3-4)

Deliverables	• Admin login API with JWT authentication • User login API for alumni members • Password encryption with bcrypt • Session management and token refresh • Protected route middleware
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Sprint 3: Frontend Foundation & Public Pages (Weeks 5-6)

Deliverables	• React Vite project setup with routing • Responsive home page with header and footer • Alumni profile card component with carousel • Registration form with validation • Login pages for admin and users
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Sprint 4: Alumni Management Backend (Weeks 7-8)

Deliverables	• CRUD APIs for alumni management • File upload API with Multer integration • Search and filter API endpoints • User management APIs • Data validation and error handling
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Sprint 5: Admin Dashboard (Weeks 9-10)

Deliverables	• Admin dashboard with statistics overview • Add new alumni form and functionality • Manage alumni table with edit/delete options • User management interface • Search and filter implementation
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Sprint 6: User Interface & Alumni Directory (Weeks 11-12)

Deliverables	• User dashboard for logged-in alumni • Alumni directory with view-only access • Advanced search and filter interface • Profile view page • Responsive design optimization
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Sprint 7: Testing, Deployment & Documentation (Weeks 13-14)

Deliverables	• Comprehensive testing (unit, integration, E2E) • Bug fixes and performance optimization • Production deployment setup • User documentation and admin guide • Training materials and handover
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10.3 Key Milestones

Milestone	Timeline	Deliverable
M1: Project Kickoff	Week 1	Project charter and requirements sign-off
M2: Backend Core	Week 4	Functional authentication and database
M3: Frontend MVP	Week 6	Working home page and registration
M4: Admin Features	Week 10	Complete admin dashboard with CRUD
M5: User Features	Week 12	Alumni directory with search functionality
M6: Go Live	Week 14	Production deployment and launch

10.4 Post-Launch Support

Following successful deployment, Sunsys Technologies Pvt Ltd will provide:

- 30 days of free post-launch support and bug fixes
- Administrator training sessions (2-3 sessions)
- Comprehensive documentation and user guides
- Email and phone support during business hours
- Optional maintenance and enhancement contract available

11. Conclusion

The Alumni Web Application proposed by Sunsys Technologies Pvt Ltd represents a comprehensive solution designed to strengthen the connection between educational institutions and their graduates. Built with modern technologies including React Vite, Node.js, and MySQL, this platform offers robust functionality for both administrators and alumni members.

Our systematic approach ensures successful delivery through well-defined sprints, clear milestones, and comprehensive testing. The platform's security features, scalable architecture, and user-friendly interface position it as a long-term asset for alumni engagement and institutional networking.

We are committed to delivering a high-quality product that meets all specified requirements while maintaining the flexibility to accommodate future enhancements. Our post-launch support ensures smooth operation and provides administrators with the knowledge and tools needed for effective alumni management.

Thank you for considering Sunsys Technologies Pvt Ltd for your Alumni Web Application project.

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